The Model Knows the Textbook but Not Your Class: Curriculum Knowledge Graph Constraints for Slide Generation in Vocational Education

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Abstract

Large language models can now produce a plausible-looking deck of lecture slides in minutes, yet the output rarely survives a teacher's review: a skill explicitly required by the syllabus is missing, the misconception that this year's cohort keeps committing is repeated rather than corrected, or material reserved for a later session is disclosed early. These failures are not defects of generation capability. They arise because the relevant information is absent from the model's parameters: course-level misconception inventories, institution-specific sequencing decisions, and the code-level landing point of a skill have never appeared in pretraining corpora.

We propose modelling such knowledge explicitly as a curriculum knowledge graph and *compiling* it, before generation, into an executable generation contract that is subsequently used, after generation, as the acceptance criterion. The central design decision is *constraint-as-data*: the contract is simultaneously the source of the prompt and the ground truth of the validator, so that what is demanded and what is checked cannot drift apart. We conduct a controlled study on the queue module of a higher-vocational *Python Data Structures* course (three sessions), covering six conditions $\times$ three resamplings, three ablations, and a length-matched placebo arm: 78 independent generations in total.

The results differ from intuition, which is why they are worth recording. Coverage rises from 0.61–0.70 at baseline to 0.98, and the length-matched

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placebo demonstrates that this is not an artefact of longer prompts. Mean misconception handling rises from 0.56 to 1.00, but the more informative change is in dispersion: the standard deviation falls from 0.53 to 0.00. A condition given the same misconception text without structure succeeds in roughly half of its generations. What the graph changes is therefore not average quality but *reliability*. On prerequisite ordering, by contrast, a strong model already performs well zero-shot, and ablation shows that the constraint contributes no reliable gain; we do not defend this negative result. We further document seven measurement defects discovered and corrected during the study, each of which silently favours the constraint-based condition and none of which raises an error—a general trap for text-level automatic metrics in educational generation.

Keywords: curriculum knowledge graph, constrained generation, slide automation, misconception modelling, educational knowledge engineering

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1. Introduction

1.1. A concrete failure

Asked to write the slides for a session on queues, a general-purpose large language model (LLM) will typically return a well-organised deck with accurate terminology and runnable code. A teacher skimming it will nod; a teacher reading it closely will frown. The failures look like this.

- The session objective states that students must *implement* enqueue and dequeue on a sequential queue using a Python list. Across the six generated slides, however, the deck never shows **how to actually write it**. It covers the definition, covers the false-overflow problem, and covers the circular queue as the remedy—everything except the step the syllabus requires students to perform.
- The session carries a misconception that more than 60% of this cohort reproduces in the final exam: "a Python `list` is a linked list underneath." The model does not mention it, because it is not a textbook topic but a property of *these* students.
- The model pushes the modular arithmetic of the circular queue into the slide on the sequential queue. Framed as "broadening the view," it in fact violates the prerequisite order: students encounter `% maxsize` before they know what `rear` is.

Table 1: Three properties of situated course knowledge and why they cannot be acquired from pretraining.

Property	Example	Why the model cannot learn it
Cohort-specific	This year's students confuse *empty* and *full* conditions at a rate above 60%	Exists only in teacher experience and assignment statistics, in no public corpus
Institution-specific	Our department merges "linked queue" and "introducing BFS" into one session	A teaching-office scheduling decision that varies by institution
Landing-point specific	This course requires "can write the code," not merely "can state the concept"	Set by the curriculum standard, not by the textbook

What these three failures share is that **none of them is a generation-capability problem**. The model is perfectly able to write this content. It simply does not know what *this course* requires, where *this cohort* errs, and how *this institution* sequences its sessions.

1.2. Thesis

We argue that the bottleneck in LLM-based slide generation has shifted from *knowledge acquisition* to the **representation and application of course-level knowledge**. A general model possesses disciplinary knowledge (what a queue is, why FIFO matters). Slide production requires *situated* knowledge: this course, this cohort, this schedule.

Situated knowledge has three properties, each of which happens to be a blind spot of current LLMs (Table 1).

None of these can be obtained through a larger model or a better prompt; they must be **supplied from outside**. The question then becomes: in what form should they be supplied, such that the model can follow them *and* the system can afterwards check whether it did?

1.3. Our answer

A **knowledge graph**. The justification is not fashion but the fact that it satisfies four requirements at once.

1. **Expressiveness.** Misconceptions, skills, prerequisite relations, and competence levels are heterogeneous entities; a graph accommodates them naturally.

2. **Orderability.** `prerequisites_for` is a directed edge, so a topological sort directly yields a legal teaching order.
3. **Compilability.** The subgraph of one session can be mechanically compiled into a generation contract (concept inventory, order, misconceptions, forbidden terms).
4. **Reusability as ground truth.** The same contract can be used to validate the output—**what is demanded and what is checked share one source, and cannot drift**.

Point 4 is the essential difference from "writing the key points into the prompt." A prompt is discarded once written, and there is no way afterwards to determine whether the model complied. A contract is a data object, readable by programs both before and after generation.

1.4. Contributions

1. **A curriculum-graph schema for vocational programming courses.** We extend the seven node types of K12-KGraph [1] by promoting `Misconception` to a first-class node, and add two edge types: `implements` (concept → executable code) and `addresses_misconception` (teaching unit → misconception). The former makes "can write it" a checkable property; the latter makes corrective content a constrainable generation target.
2. **A constraint-as-data generation framework.** `UnitConstraints` drives both prompt construction and post-hoc validation; topological order, required/optional partitioning, sibling-session background caps, and forbidden-term lists are all derived mechanically from the subgraph, with no hand-written rules.
3. **A reproducible six-metric acceptance protocol**, two of whose metrics are redefined for this task: prerequisite order is judged by *where a concept is named in a slide title* (not by model-reported IDs), and coverage is matched against *graph synonyms* (so that a condition copying the source wording gains no advantage).
4. **An honest empirical study.** Six conditions × three sessions × three resamplings, plus three ablations and a length-matched placebo: 78 independent generations. We report gains *and* the absence of gain: misconception handling differs decisively in both mean and variance; coverage improves by about 0.3 with the length explanation excluded by the placebo; and ordering constraints show no reliable contribution un-

der a strong model. We also document **seven** measurement defects that systematically favour constraint-based methods, together with their fixes (§7.1).

1.5. Why this belongs in Knowledge-Based Systems

The object of this paper is not "a better slide generator" but **how knowledge must be represented in order to be actionable by a machine**: what structure suffices to constrain generation, how it is compiled, how it is verified, and along which dimensions a knowledge base is genuinely irreplaceable. This matches the knowledge-representation, reasoning and system-evaluation loop that this journal serves.

Three recent contributions in this journal form the immediate context. Dai et al. [2] systematically compare how LLMs comprehend knowledge graphs presented in different prompt formats (linearised triples, fluent natural language, neighbourhood expansion), showing that **the presentation format itself changes performance**. Huang et al. [3] find that graph-triple prompting generally outperforms passage prompting on long-tail factual question answering, and that combining them markedly reduces hallucination, showing that **structure and text are not interchangeable**. Ma et al. [4] apply knowledge-enhanced LLMs to course recommendation, placing education among the journal's active application areas. We continue this line but move the question from *how to put knowledge in* to *whether the knowledge once inserted can also serve as the acceptance criterion.*

We emphasise **negative results**: a knowledge graph is not superior to a strong model along every dimension, and identifying *which* knowledge must be externalised is more valuable than a blanket claim that knowledge augmentation works.

2. Related work

2.1. Automatic slide generation

Three strands can be distinguished, each solving a different problem.

Document-to-slide: fidelity to the input. DOC2PPT [5] first formalised the task, observing that it simultaneously demands document summarisation, figure–text retrieval and layout arrangement. It proposed a hierarchical sequence-to-sequence model and released 5,873 paired document–slide examples drawn from computer-vision, ACL and ICML proceedings, together with

four metrics (slide-level ROUGE, LC-FS, TFR, mIoU). Mondal et al. [6] introduced *end-user specifications*: the same paper should yield different decks for different audiences or durations. They built four persona configurations on SciDuet, fine-tuned an LLM on target-audience feedback, and cut preparation time from about one hour to under thirty minutes. Notably, the authors state in their limitations that they **do not address slide layout design**; the contribution is content and logical alignment only.

Instruction-to-slide: fidelity to editability. AutoPresent [7] released SlidesBench (7k training and 585 test examples from 310 real decks across ten domains) and compared end-to-end image generation against program generation, concluding that **programmatic generation yields higher-quality slides** because the artefacts are interactive and re-editable; an 8B model trained on the benchmark matches GPT-4o. PPTAgent [8] adopts a two-stage edit-based approach: analyse reference decks to extract slide-level functional types and content schemas, then draft an outline and iteratively generate editing actions, with PPTEval assessing content, design and coherence. Our rendering decisions (native shapes, a mandatory JSON intermediate representation) follow this strand's conclusions.

Evaluation: from "does it look right" to "is it done". PPTC [9] changes the question from whether slides look good to whether a model can modify a deck correctly under multi-turn instructions. Its 279 multi-turn sessions come with PPTX-Match, which compares the resulting `.pptx` file state rather than the API sequence and therefore admits multiple valid solutions. GPT-4 reaches 75.1% single-turn accuracy but only 6% session accuracy, with errors traced to multi-turn accumulation, long template processing and multimodal perception.

Our difference. The inputs to the above work are a *document to be compressed* or *a single instruction*; the criteria are resemblance to the source, visual appeal, or task completion. Our input is a *course specification* (what to teach, in what order, which misconception to correct, what not to introduce), and the criterion is conformance to the syllabus. This is not a relabelling: neither the reference-based nor the reference-free protocol of SlidesBench can express "the deck omits the skill the syllabus requires students to implement," which is precisely the most damaging class of failure in a teaching context.

2.2. Educational knowledge graphs

K12-KGraph [1] is the closest knowledge-side work. Extracted from the official People's Education Press textbooks, it defines seven node types (Book, Chapter, Section, Concept, Skill, Experiment, Exercise) and nine relation types (`is_a`, `prerequisites_for`, `relates_to`, `verifies`, `tests_concept`, `tests_skill`, `appears_in`, `leads_to`, `is_part_of`), covers 48 volumes across mathematics, physics, chemistry and biology, and yields 10,685 nodes and 23,278 edges. From it the authors derive K12-Bench (23,640 multiple-choice items in five task families: Ground, Prereq, Neighbor, Evidence, Locate) and K12-Train, a graph-guided supervised fine-tuning corpus. Its most telling number is not the overall score but the fact that **even the strongest proprietary model leaves more than 40% of Prereq and Neighbor items unsolved**—evidence that structural curriculum cognition is a capability dimension independent of factual recall, and precisely the dimension we attempt to supply externally.

SciMKG [10] moves to multimodality: an LLM-driven extraction–verification–integration–augmentation pipeline builds a graph over text, images, video and audio from open courseware, yielding 1,356 knowledge points, 34,630 multimodal concepts and 403,400 triples, with concept-extraction F1 improved by 9% over baselines. On the application side, Ma et al. [4] apply knowledge-enhanced LLMs to course recommendation, confirming sustained interest in the combination of knowledge structure and education.

The centre of gravity in this literature is *building* graphs (extracting entities and relations from corpora, ensuring consistency). Ours is *using* them (compiling a graph into generation constraints and validating against it). The two are complementary: our method can consume graphs produced by the above work directly, and indeed we reuse the K12-KGraph node and relation skeleton, adding to it rather than replacing it (§4.1).

In vocational education, a graph must additionally express two things that general subject-matter graphs do not emphasise: **competence level** (understand, comprehend, master, proficient, which decides whether a concept is mandatory or optional) and **typical misconceptions** (which decide corrective content). K12-KGraph carries an `importance` attribute on concepts but does not use it to constrain generation; misconceptions are absent entirely. This is the direct motivation for our schema extension.

2.3. Constrained generation

Classical natural language generation separates *what to say and in what order* (content planning) from *how to say it* (surface realisation) [11]. Wiseman et al. [12] showed on RotoWire that end-to-end models produce fluent game reports while systematically erring on content selection and discourse order, in some metrics even being surpassed by template baselines. Puduppully et al. [13] therefore model content planning explicitly: first predict which records to mention and in what order, then generate conditioned on that plan. What unites this line is that structure is injected into the *training objective or the decoding process*.

In the LLM era, structure is injected through prompts instead, and the question becomes *in what form*. Dai et al. [2] provide the most systematic evidence to date: the same graph, presented as linearised triples, fluent paragraphs or expanded neighbourhoods, is comprehended with markedly different effectiveness, and the difference is explicable from attention distributions. Huang et al. [3] offer a complementary result on long-tail factual question answering: triple-based prompting generally beats passage-based prompting, and combining both does not always improve coverage but does markedly reduce hallucination. In short, **structured knowledge is not simply "text written differently."**

We differ from both lines deliberately: we neither train, nor modify decoding, nor rerank. We compile constraints into an *executable data object*, use it to construct the prompt before generation, and use the same object to validate afterwards. Relative to [2, 3], the question advances from "how should a graph enter the prompt" to "can one graph simultaneously serve as the acceptance criterion"—if so, demand and check cannot drift (§1.3). The price is that we relinquish control over decoding and must bear the risk that the model disobeys; §5.5 reports a real violation and §7.1 reports seven metric traps that this permissiveness exposed.

3. Problem formulation

3.1. Curriculum knowledge graph

A curriculum knowledge graph is a typed directed heterogeneous graph $G = (V, E, \tau)$, where $\tau : V \to T_V$ with

$$T_V = \{\text{Course}, \text{Module}, \text{Unit}, \text{Concept}, \text{Skill}, \text{Lab}, \text{Exercise}, \text{Misconception}\};$$

each edge $e = (u, r, v) \in E$ carries a relation type $r \in T_E$ with

$$T_E = \{\text{is_a},\ \text{prerequisites_for},\ \text{relates_to},\ \text{implements},\ \text{verified_by},\ \text{assessed_by},\ \text{addresses_misconception},\ \text{appears_in},\ \text{is_part_of},\ \text{leads_to}\};$$

and selected edges carry an `evidence` field recording the provenance of the relation (textbook page, curriculum-standard clause, error-rate statistic).

Constraint 1 (acyclicity). The subgraph induced by `prerequisites_for` must be a DAG. We detect violations with Kahn's algorithm: if the concept set of a session cannot be topologically ordered, the course design itself contains a circular prerequisite, and construction fails immediately rather than deferring the contradiction to generation time.

3.2. Session contract

For a session $u \in V$ with $\tau(u) = \text{Unit}$, take the subgraph

$$G_u = \{\, v \in V : (v, \text{appears_in}, u) \in E \,\} \tag{1}$$

and compile it into a contract

$$C(u) = \langle S_u,\ \prec_u,\ M_u,\ F_u,\ B_u,\ k_u \rangle, \tag{2}$$

whose components are given in Table 2.

A critical distinction separates B_u from F_u. The former means "already covered, may be mentioned in one sentence"; the latter means "not taught in this course at all." Conflating them makes two metrics fight each other (the same term counts both as legitimate recap and as out-of-scope), so the implementation enforces mutual exclusivity by assertion.

Compilation is mechanical and lossless. Fig. 2 shows the actual U11 subgraph and the contract compiled from it: every edge in panel (a) maps to a field in panel (b), and nothing in panel (b) is hand-written.

3.3. Generation and acceptance

Generation is a mapping $D = f_\theta(C(u))$ where D is a sequence of slides. Acceptance is a metric vector

$$m(D, C(u), G) = \langle \text{cov},\ \text{oog},\ \text{bg},\ \text{pre},\ \text{mc},\ \text{edit} \rangle \tag{3}$$

Table 2: Components of the session contract $C(u)$ and where each is read from in the graph.

Symbol	Meaning	Source
$S_u = (R_u, O_u)$	required / optional concept sets	split on `importance`; at or above "master" is required
$\prec_u$	teaching order (a total order on S_u)	topological order of `prerequisites_for` over G_u
M_u	misconceptions that must be corrected	`Misconception` nodes targeted by `addresses_misconception`
F_u	forbidden-term list	`forbidden_terms` on the Unit node
B_u	sibling-session concepts admissible as brief recap	concepts of other Units in the same Module, deduplicated
k_u	slide budget	`slide_budget` on the Unit node

defined in §5.3. Note that $C(u)$ appears on both sides: this is the formal statement of "demand and check share one source."

The full pipeline is shown in Fig. 1: graph → contract compilation → generation → rendering → validation against the same contract. One loop exists: pages that fail validation are sent back for regeneration. Its boundary is deliberately fixed—**the loop may change slides, but never the graph and never the contract**—and whether to regenerate is a teacher's decision, not an automatic optimisation. Were validation output allowed to reshape the contract, acceptance would degrade into an objective and the metrics would lose their meaning (§7.1).

4. Method

4.1. Graph construction

4.1.1. Nodes and edges

We retain the seven node types of K12-KGraph [1] (`Lab` corresponds to its `Experiment`) and promote `Misconception` to an eighth, with attributes `statement` (the erroneous claim itself, which must be concrete and decidable), `correction`, `frequency` (low, common, high) and `evidence` (exam error rate, assignment statistics, in-class questioning).

Two edge types are added:

- `implements`: Concept or Skill → executable code fragment. It makes "can write it" a checkable property and supplies the generator with runnable examples.

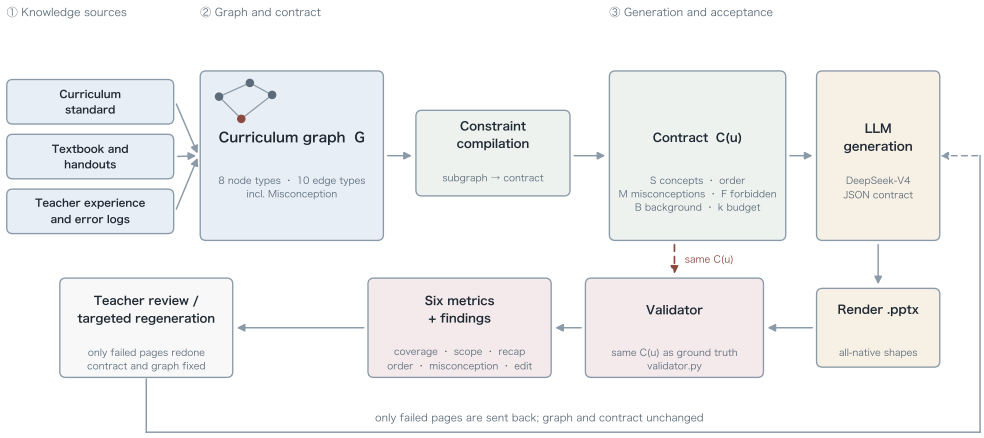


Figure 1: Framework overview. The contract $C(u)$ is read once before generation (to build the prompt) and once after (to compute the six metrics); both reads concern the same object, which is what "demand and check share one source" means in implementation terms.

- `addresses_misconception`: Unit, Lab or Concept → Misconception, annotated with a strategy (counter-example, code demonstration, boundary test). This edge decides **which session is responsible** for correcting a given misconception, preventing both duplication across sessions and orphaned misconceptions.

4.1.2. Synonyms

Every concept may carry aliases. This is not a patch for evaluation but legitimate graph content: in class the teacher says "judging full," the textbook writes "the full condition," and the code reads `(rear+1) % maxsize == front`; all three denote one entity. Without aliases the graph cannot be aligned with real teaching language. Aliases also resolve a fairness problem in evaluation, as discussed in §7.1.

4.2. Constraint compilation

`build_constraints` turns the subgraph into a `UnitConstraints` object with no hand-written rules:

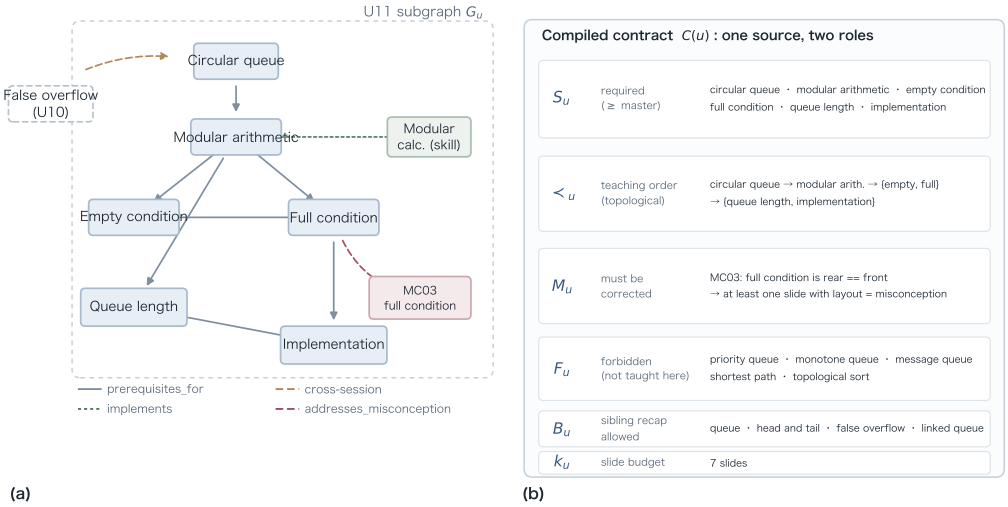


Figure 2: From subgraph to contract. (a) The actual U11 subgraph in the released graph file: six concepts and one skill, seven intra-session prerequisite edges, one cross-session prerequisite edge from U10, one `implements` edge and one `addresses_misconception` edge. (b) The contract $C(u)$ produced from it by `build_constraints`; the validator subsequently reads the same object.

1. **Required/optional split.** Sort by `importance`; those at or above "master" enter R_u, the rest O_u.
2. **Topological order.** Run Kahn's algorithm over S_u; on encountering a cycle, fall back to declaration order and raise a warning.
3. **Misconception assignment.** Collect the misconception nodes attached by `addresses_misconception` edges.
4. **Sibling sessions.** Concept names of other Units in the same Module, minus those already in this session.
5. **Evidence anchors.** Collect `evidence` along incoming `prerequisites_for`, `implements` and `verified_by` edges, and render them as *teacher-facing* verification material, explicitly labelled "not generated content" so that the model does not copy the provenance into the slides as body text.

The compilation output is a *deterministic string*: the same graph always yields the same prompt block, which is a precondition for reproducibility.

4.3. Generation

The prompt is assembled from four segments—a fixed skeleton, the contract block, a JSON contract, and a list of hard rules—and the experimental conditions of §5.2 replace only some of them.

Output is constrained to a single JSON object, each slide carrying `layout` (one of seven), `title`, `bullets`, `code`, `notes`, `concept_ids` and `background`. Forcing JSON rather than free-form layout is a deliberate choice: layout selection is itself an instructional-design decision, and delegating it to the model makes both editability and slide budget uncontrollable.

Two engineering realities, both observed rather than assumed:

1. **JSON mode is not a validity guarantee.** Roughly once every ten calls the response contains a trailing comma, or returns blank while reporting `finish_reason == "stop"` (the offending fragment replaced by whitespace). The pipeline therefore implements *call* → *conservative repair* (strip trailing commas, close truncated brackets) → *parse*, and only escalates by re-prompting with the concrete parse error when repair fails. Blind retries reproduce the same error at the same cost.
2. **The model performs internal reasoning by default.** Every call in our setup produced `reasoning_tokens`, which contaminates a chain-of-thought baseline (the model is already doing CoT). We record the setting explicitly and provide a switch to disable it for a robustness check.

4.4. Rendering

`python-pptx` emits 16:9 fully native shapes with no rasterised text. `editability` is defined as the ratio of native editable shapes to all shapes; it is 1.0 for every condition here. This dimension does not discriminate between methods; it exists to exclude "screenshot-style generation," whose value for a teacher who needs to edit is zero.

For Chinese text, the East-Asian and complex-script typeface attributes must be set explicitly, otherwise presentation software falls back to a Latin font and glyphs are corrupted.

4.5. Validation

The six metrics are defined in §5.3. Three implementation choices are deliberate.

- **Chinese has no word boundaries.** The term for *queue* matches inside the term for *circular queue*. All counting therefore uses per-line net occurrences: raw count minus occurrences covered by longer terms *within the same line*. Subtracting over the whole document rather than per line is a real bug: an independent occurrence on slide 2 is otherwise cancelled by a compound term on slide 4.
- **Triple-criterion misconception detection.** Structural anchor (does the misconception slide reference a graph-linked concept), then English-identifier overlap (`rear`, `list`), then Chinese character bigram overlap (threshold 0.35). Purely lexical matching misses cases where the model paraphrases the erroneous claim in its own words; we observed a generated correction slide rejected because its phrasing differed from the graph's.
- **Order by first textual position**, see §5.3.

5. Experiments

5.1. Course and data

The target course is *Python Data Structures*, a higher-vocational programme in Artificial Intelligence Technology Application with 136 contact hours. The graph scope is Module M4, "Queues", comprising three sessions (U10, concepts and false overflow; U11, circular queue; U12, linked queue and introducing breadth-first search), with 34 nodes, 75 edges and four misconceptions. The prerequisite chain in this module is clean (queue $\rightarrow$ sequential queue $\rightarrow$ circular queue $\rightarrow$ linked queue), whereas deep-learning courses have mutually referencing concepts that easily form cycles and are therefore unsuitable as a first validation scenario.

The goal is to establish *whether and along which dimensions* graph constraints help, not to maximise a course-wide number. Three sessions cover three qualitatively different cases: concept-dense (U11 has seven required concepts), skill-landing (U10 contains a skill that must be implemented by hand), and application-introduction (U12 requires a comparative selection against prior material).

5.2. Conditions

Six conditions carry strictly increasing information, each step removing one structural element (Table 3).

- `rag` − `zero_shot` measures the contribution of *content retrieval*.

Table 3: Six experimental conditions. Prompt lengths are measured on session U11, in characters.

Condition	Course/ session	Page budget	Concept names + definitions	Topological order	Misconceptions	Forbidden terms
`full` (ours)	✓	✓	✓	✓	✓	✓
`rag_mc`	✓	✓	✓	–	✓	–
`rag`	✓	✓	✓	–	–	–
`placebo`	✓	✓	–	–	–	–
`cot`	✓	✓	–	–	–	–
`zero_shot`	✓	✓	–	–	–	–
Prompt length (U11)				1997 / 1122 / 997 / 2195 / 625 / 585		

- `full` − `rag` measures the contribution of *structural constraints*.
- `rag_mc` is a control built specifically for the misconception claim: the same misconception text is supplied as a flat passage, but without ordering and without layout requirements. If `full` still wins on the misconception metric, the gain comes from *structuring the text to the point where it constrains layout*, not from *the text appearing in the prompt*.
- `placebo` is a control built specifically for the alternative explanation "the prompt is simply longer": the prompt is *length-matched* to `full` (in fact slightly longer), but padded with generic slide-writing advice unrelated to this course. If the advantage of `full` were prompt length, the placebo should catch up. It does not (§5.4).

Three ablations retain the complete contract and rule wording of `full`, deleting exactly one data block (Table 4).

The rule text must be removed together with the data. Deleting only the data while keeping a rule that says "you must follow the given order" forces the model to guess an order, and the ablation is no longer clean; a unit test guards this.

5.3. Metric definitions

The six metrics are listed in Table 5.

Table 4: Three ablations. The corresponding rule text is removed together with the data block.

Condition	Removed	Rule also removed	Expected metric
`no_order`	order block	rule 3	`prerequisite_violation`
`no_misconception`	misconception block	rule 5	`misconception_coverage`
`no_forbidden`	forbidden-term block	rule 6	`oog_rate`

Table 5: Metric definitions. Arrows indicate the desired direction.

Metric	Definition	Direction
`coverage_rate`	fraction of required concepts actually taught	↑
`oog_rate`	(out-of-graph IDs + forbidden-term hits) / (covered + out-of-graph)	↓
`background_rate`	characters devoted to sibling-session concepts / total characters (cap 25%)	↓
`prerequisite_violation`	violated concept pairs / all concept pairs	↓
`misconception_coverage`	misconceptions corrected / misconceptions linked to the session	↑
`editability`	native editable shapes / all shapes	↑

Three definitional decisions determine the conclusions and therefore require explicit justification:

1. **Coverage counts body slides only.** Title, objectives and summary slides list every concept by definition; counting them as "taught" lets a deck consisting of a table of contents score full marks.
2. **Coverage matches on name and aliases.** Otherwise the `full` condition, which copies source names, enjoys a structural advantage (§7.1).
3. **Order is judged by first textual position.** The model's `concept_ids` are self-reported, and the three baselines were never told the field exists. Scoring on self-reports means a completely empty answer receives a zero violation rate.

Table 6: Main results ($n = 9$), mean (standard deviation). Cov.: coverage; OOG: out-of-graph; Bg.: background; Ord.: prerequisite violation; Miscon.: misconception. Best value per column in bold. Editability is 1.000 for every condition and is therefore omitted.

Condition	Cov. ↑	OOG ↓	Bg. ↓	Ord. ↓	Miscon. ↑	Pass rate
`full` **(ours)**	**0.981** (0.056)	**0.000**	0.027 (0.016)	**0.020** (0.045)	**1.000** (0.000)	67%
`rag_mc`	0.698 (0.295)	0.111 (0.220)	0.027 (0.016)	0.007 (0.022)	0.556 (0.527)	22%
`rag`	0.659 (0.248)	0.111 (0.220)	0.032 (0.014)	0.030 (0.089)	0.222 (0.441)	0%
`placebo`	0.675 (0.266)	0.167 (0.250)	0.040 (0.018)	0.096 (0.131)	0.333 (0.500)	0%
`cot`	0.614 (0.185)	0.019 (0.056)	0.040 (0.016)	0.089 (0.183)	0.389 (0.486)	0%
`zero_shot`	0.643 (0.248)	0.019 (0.056)	0.040 (0.011)	0.037 (0.089)	0.222 (0.441)	0%

5.4. Results

Each condition runs three sessions × three resamplings, that is, nine independent generations; each ablation runs six. Values are mean (standard deviation).

Four observations follow, ordered by importance.

(1) Coverage: 0.981 against 0.61–0.70, and not an effect of prompt length. The five controls lie within 0.09 of one another (0.614–0.698) while all have large dispersion (0.19–0.30); `full` is about 0.3 higher with dispersion down to 0.056. The decisive evidence is the placebo: its prompt is comparable in length to `full` (2195 against 1997 characters on U11, in fact longer), but its content is generic writing advice unrelated to the course, and its coverage is 0.675, indistinguishable from zero-shot. The coverage gain therefore comes from **course structure**, not from saying more.

(2) Misconception: the mean is the lesser finding; the disappearance of variance is the greater. This is the strongest evidence in the paper. `rag_mc` gives the model the same misconception text and reaches a mean of 0.556, but with a standard deviation of 0.527: it succeeds reliably in only about half of nine generations. `full` reaches 1.000 with zero dispersion. In other words, **telling the model the knowledge yields a system that depends on luck; structuring the knowledge into a layout requirement yields a reliable system.** This is precisely our thesis: the value of the graph is not that it conveys content but that it forces content to appear *at a specific position.*

Table 7: Ablations; each removes exactly one part of the graph ($n = 6$).

Condition	Coverage ↑	Out-of-graph ↓	Order viol. ↓	Misconception ↑
`full`	1.000 (0.000)	0.000 (0.000)	0.032 (0.078)	1.000 (0.000)
− ordering	1.000 (0.000)	0.000 (0.000)	0.000 (0.000)	1.000 (0.000)
− misconceptions	0.944 (0.086)	0.000 (0.000)	0.040 (0.064)	**0.167** (0.408)
− forbidden terms	1.000 (0.000)	**0.083** (0.129)	0.000 (0.000)	1.000 (0.000)

(3) Ordering: the constraint helps but not significantly, and adds no reliable marginal contribution. The violation rate of `full` (0.020) is below that of the placebo (0.096) and `cot` (0.089), but `zero_shot` sits at 0.037: a condition given no ordering information at all is nearly as good. This is consistent with the ablation below and with the explanation that strong models already have a sound intuition for textbook prerequisite chains. We do not defend it: **on this task, ordering is not where the knowledge graph earns its keep.**

(4) Out-of-graph: `rag` and `rag_mc` are worse than zero-shot. Both reach an out-of-graph rate of 0.111 and the placebo 0.167, whereas `zero_shot` sits at 0.019 and `full` at 0. Telling the model "this session covers these points" leads it to bring in adjacent advanced material (mentioning priority queues in a session on queues), because the prompt demarcates a scope without demarcating a boundary. `full` reaches zero only by virtue of the forbidden-term list, an explicit negative example, as the ablation confirms.

The ablation (Table 7) yields two positives and one negative, cleanly separated:

- Removing the misconception block drops misconception coverage from 1.000 to 0.167 and coverage from 1.000 to 0.944; the block carries corrective content and incidentally helps the model organise the session's key points.
- Removing the forbidden-term list raises the out-of-graph rate from 0.000 to 0.083 with every other metric unchanged: single responsibility, clean attribution.
- Removing the ordering constraint drops the violation rate from 0.032 to **0.000**, opposite to expectation. This is not a measurement error: `full`'s 0.032 traces to one genuine inversion in U10 (sequential queue taught before head and tail). At the scale of seven concepts and 21 concept pairs, a single inversion is of order 0.05, so the difference lies entirely within noise. **We conclude that the ordering constraint has no reliable**

Table 8: Cost and latency per generation.

Condition	Input tok	Output tok	of which reasoning	Latency (s)
`full`	1064	4079	2708	20.8
`rag_mc`	618	2689	1441	14.4
`rag`	536	2378	1133	12.3
`cot`	373	2638	1157	14.4
`zero_shot`	347	2330	984	12.4

effect on this data.

`full` emits about 1.75 times the output tokens of `zero_shot`, with end-to-end latency of 20.8 s against 12.4 s (Table 8). The surplus comprises the contract block itself (about 500 input tokens) and longer slide content with more reasoning tokens. Per deck the cost remains in the "cents" range, negligible against teacher preparation time; for course-wide batch generation, however, the 1.75× gap is worth engineering away, for example by caching contract blocks per session.

Note that the longer output of `full` is not due to more slides—the page count is fixed by `slide_budget` and met in every run—but to fuller speaker notes on each slide. This is a benefit for teachers, since speaker notes are the bulk of lesson preparation.

5.5. Qualitative observations

A positive case: `full` on U12. U12 is the hardest of the three sessions: it requires students to *choose* between a circular and a linked queue, a judgement task with no single correct answer, and one of its required concepts ("application scenarios of queues") is abstract and easily reduced to platitudes. All three `full` runs reached coverage 1.000 and each produced a genuine comparison slide (predictable capacity and frequent queries favour the circular queue; highly variable length and frequent enqueue–dequeue favour the linked queue), handling the breadth-first-search misconception on a dedicated counter-example slide. Coverage for the five baselines on the same session is **0.333**: they cover only "linked queue" and dispose of selection and scenarios in a single clause.

A negative case: `full` fails too. In the third resampling of U10, coverage is 0.833, missing the concept "queue" itself. Reading slide by slide: the title slide states that a queue is a linear list inserting at one end and deleting

Table 9: Contribution of the knowledge graph, by dimension.

Dimension	Verdict	Evidence
Misconception handling	Irreplaceable	`full` 1.000 ± 0.000 against `rag_mc` 0.556 ± 0.527 with the same text
Coverage completeness	Irreplaceable	0.981 against 0.61–0.70; length-matched placebo excludes prompt length
Scope control	Partial	Rests on the forbidden-term list as an explicit negative example; ablation confirms (0.000 → 0.083)
Prerequisite order	**No reliable contribution**	`full` 0.020 against `zero_shot` 0.037; ablation runs in the opposite direction
Editability	Orthogonal	1.000 everywhere; determined by the renderer

at the other, but the first body slide is titled "head, tail and the sequential queue," and the deck never defines a queue separately again. This is not a metric artefact—the definition genuinely stayed on the title slide, with the model assuming students would remember it. The validator flagged it as a missing concept, which is exactly the point of closing the loop: the method does not guarantee correctness, it guarantees that **errors are not silently passed over**.

A second failure is more instructive still: the second resampling of U10 contains two prerequisite inversions, the model teaching the sequential queue first and returning to head and tail afterwards. On the metrics this is a 0.133 violation rate; in the classroom it means students first encounter the head and tail indices without knowing what they denote. The topological order was in the prompt and the model violated it anyway, which is why we insist on post-hoc validation rather than assuming that saying it implies doing it.

6. Discussion

6.1. Where the graph helps and where it does not

Rearranging the six metrics by the question "does the graph make an irreplaceable contribution" gives a clear picture (Table 9).

This ordering is itself a result: **the graph supplies not disciplinary knowledge but the layer of institutional convention around it.** Disciplinary knowledge—what a queue is, why FIFO matters, what to teach first—is already handled well enough by a strong model, so the marginal value of the graph along the ordering dimension is close to zero. Conversely,

which concepts count as "must be finished in this session," where this cohort errs, and what belongs to the next session appear in no pretraining corpus; once the graph supplies them, the model uses them, and uses them stably.

6.2. Three kinds of knowledge that must be externalised

Returning to the taxonomy of §1.2, the experiments assign each a strength of evidence.

(1) Cohort-specific misconceptions: strongest evidence. This is the only dimension where "given the text, still unstable; given the structure, stable" holds. The half-success rate of `rag_mc` shows that, having seen the misconception, the model writes a correction slide in about half the cases and folds it into a bullet elsewhere in the other half, after which it disappears. `full` converts it into a hard constraint through a layout-level requirement of at least one slide with a misconception layout. **The conclusion: misconception knowledge must not only be stated, it must be stated into position.**

(2) Skill landing points from the syllabus: clear evidence. Baselines systematically omit skills such as implementing enqueue and dequeue on a sequential queue using a list. Not because the model cannot write code, but because it does not know that this session requires *writing* rather than merely *understanding*. Competence level is fixed by the curriculum standard and cannot be inferred. The 0.98 against 0.67 coverage gap comes largely from here: on U12, 0.333 against 1.000.

(3) Institution-specific prerequisite conventions: insufficient evidence. We expected this to be a key selling point; it does not hold empirically. Strong models already have a stable intuition for textbook prerequisite chains, and this course's order within the module matches the standard textbook, leaving no discriminating power. Testing this claim requires a test set built on a syllabus that *deliberately departs* from the standard order. This is the most worthwhile follow-up.

6.3. A measurement lesson worth remembering

Perhaps the most valuable sentence in this paper is not that coverage improved by 0.3 but the following.

> On the misconception dimension, the mean of `rag_mc` differs from that of `full` by only 0.44, while its standard deviation differs by 0.53. What the graph changes is not *average quality* but the *reliability of quality*.

When a generative system enters a teaching workflow, what teachers care about is rarely "how good can it get" but "how bad can it get," because a class happens on schedule and one failure means redoing the work. By that standard, a 0.556 ± 0.527 system is unusable and a 1.000 ± 0.000 system is usable. This also suggests that evaluation in this area should report *variance*, not only means; in the comparable literature we surveyed, variance is frequently absent.

6.4. The value of negative results

If a knowledge-augmentation method wins on every metric, the usual explanation is that the baseline was weakened. That we report no gain from ordering, no discrimination from editability, and a 33% non-pass rate for `full` itself is a credibility signal: the metrics do not systematically favour the constraint-based condition (we fixed seven defects to that end, §7.1), and the value of the knowledge base is concentrated in a narrow but critical gap. For readers of this journal the gap is worth noting: it cannot be closed by a better generator, because it is not a capability problem but a problem of information provenance.

7. Threats to validity

7.1. Text-level metrics systematically favour constraint-based methods

This is the most important methodological caution in the paper. During the study we discovered and corrected **seven** measurement defects (Table 10). Each made the `full` condition win without cause, and not one raised an error: the metrics kept emitting plausible numbers that happened to be wrong. None was found by reflection; all came from printing model outputs slide by slide and reading them.

The common structure across all seven is that **the representation the constrained method receives is exactly the representation the metric uses.** Any metric that scores against the list in the prompt encounters this problem.

Table 10: Seven measurement defects that silently favour the constraint-based condition, and their fixes.

#	Defect	Symptom	Consequence	Fix
1	Verbatim coverage matching	Graph says "enqueue and dequeue operations", baseline writes "enqueue and dequeue"	Baseline coverage crushed to 0.50 while `full` scores 1.00 by copying source wording	Add aliases to the graph; one shared vocabulary across conditions
2	Order scored from self-reported IDs	Baselines never fill `concept_ids`	Empty sequence gives 0.00 violation; **an unanswered script scores full marks**	Score from textual position instead
3	Preview slides counted as body	Title and objectives slides list every concept	Ordering determined by list order rather than teaching order; three spurious violations in U10	Exclude title, objectives and summary
4	Forward reference treated as teaching	One slide remarks that modular arithmetic is key to implementing the circular queue	The concept is located three slides early; two spurious violations in U11	Locate by title naming first, fall back to first mention
5	Cross-slide subtraction	The simple term is cancelled by a compound term on another slide	Concept judged "never taught"; coverage falsely reported as 0.83 in U10	Compute net occurrences per line
6	Misconception anchors too broad	Unit-level `addresses_misconception` used as a slide anchor	Any slide with the same layout counts as corrected; the ablation still scores 1.00	Restrict anchors to concept level; fix one mis-attached data edge
7	Out-of-graph test too strict	Baselines invent their own ID scheme	Out-of-graph rate inflated to 0.67 when the only difference is a namespace	Count only non-existent IDs matching the graph's ID format

Defect 6 deserves separate mention, because it is not a bug in the metric code but **an error in the graph data**. We attached an edge from a misconception about breadth-first search to a concept about empty-queue detection in the linked queue, on the reasoning that they were "related." They are not, instructionally. The consequence was that any misconception-layout slide referencing empty-queue detection counted as having addressed the other misconception, so the ablation still scored full marks and the entire conclusion about misconception injection nearly became false. **Graph errors silently contaminate metrics**, a risk specific to knowledge-graph-driven methods that purely prompt-based methods do not face.

We therefore recommend that work of this kind report two things: (a) how the metrics behave *on the baselines*, not only the final numbers; and (b)

at least one manual pass in which outputs are printed slide by slide, rather than reading only the summary table.

7.2. Sample and external validity

Three sessions, one course, one model, two to three resamplings per cell. Statistically this supports descriptive reporting only; it does not license significance claims. Generalisation across courses (especially non-programming ones) and across models (including open-weight ones) is untested.

7.3. Absence of human grading

The six metrics measure *conformance to the contract*, not teaching quality. Whether examples are apt, pacing sensible, or board work clear is entirely beyond them. Blind teacher evaluation is a necessary complement and is not yet complete.

7.4. The constructedness of the misconception result

`misconception_coverage` is structurally unfavourable to baselines, since they were never told about the misconceptions. Our response is the `rag_mc` control, which supplies the same text in flat form. If `full` still wins, the gain comes from structure rather than information. Even so, we must concede that the metric measures *whether* the course-required correction was made, not *whether it is correct*.

8. Conclusion and future work

We reframe slide generation as a **course-specification satisfaction problem**, carry situated knowledge in a knowledge graph, and use constraint-as-data so that one contract serves simultaneously as generation basis and acceptance ground truth. Empirically, a strong model is already good enough at prerequisite ordering; what the graph makes irreplaceable are *the skill landing points in the syllabus* and *the cohort-specific misconceptions*, neither of which exists in any pretraining corpus.

Future work proceeds along four lines: (i) extend to all sessions of the course and introduce blind teacher evaluation; (ii) build graphs semi-automatically from textbook PDFs to reduce authoring cost; (iii) add a render-and-review loop (screenshot, detect overflow and overlap, repair automatically); and (iv) release a Chinese vocational slide-generation benchmark so that comparable work can be compared on common ground.

CRediT authorship contribution statement

Suxing Liu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The curriculum graph, the generation pipeline, the experimental protocol and all raw model outputs are openly available in the `kg-slidegen` repository. Each experimental cell retains three artefacts: the rendered deck, the raw model output, and an itemised validation report.

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Appendix A. Reproducibility

Code, graph, experimental protocol and all raw outputs are available in the `kg-slidegen` repository. The accompanying protocol document records how every number in this paper was produced, and each experimental cell retains three artefacts: the presentation file, the raw model output and a per-item validation report.